

MedVoice AI

Product Architecture & Pre-Code Plan

Complete Product Guide

June 2026

Abstract

What MedVoice is, how it works today, clinic vs. super-admin dashboards, database design, and the full step-by-step checklist before writing more code.

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1 What This Product Is

1.1 One-Sentence Pitch

MedVoice AI is a 24/7 AI medical receptionist sold to clinics as a subscription. It answers phone calls and website chats, collects patient information, books appointments into Google Calendar, escalates emergencies, and surfaces everything in a clinic dashboard.

1.2 The Analogy

You sell two things:

1. **The product** — prompts, medical safety, booking logic, dashboard, billing. *This is MedVoice.*
2. **The voice engine** — real-time phone audio via Vapi + Twilio. *Rented infrastructure.*

Clinics care that calls get answered and appointments land on their calendar — not which runtime you use.

1.3 Who Buys It

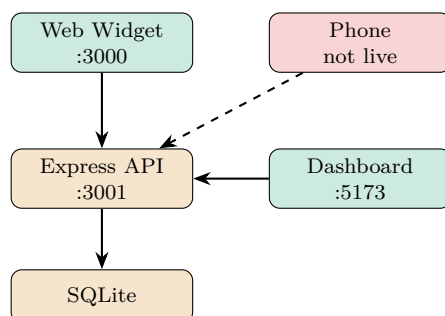
Buyer	Pain
Solo doctor / small clinic	Missed after-hours calls; no budget for 2 FTE staff
Dental / urgent care	High volume; repetitive booking questions
Multi-provider group	Inconsistent front desk; staff turnover

1.4 Problems Solved

- Missed calls → lost appointments (\$150–300 each)
- After-hours voicemail → patients call the next clinic
- Staff cost: ~\$9K/mo (2 receptionists) vs. \$1.5–3K/mo (MedVoice)
- Emergency risk: hard-coded detector bypasses LLM for red-flag keywords

2 How It Works Today

2.1 Three Apps in the Repo



2.2 Web Chat Flow (Functional)

1. Patient opens widget (`?client=medical-clinic`)
2. Widget loads config from `GET /api/config`
3. **ResponseOrchestrator** runs 6-step pipeline per message

4. Emergency detector runs first (keyword match, bypasses LLM)
5. Lead capture collects name, DOB, insurance, reason
6. Turns logged to SQLite; webhooks fire to clinic n8n URL
7. Clinic staff views data in dashboard

2.3 What Is Not Working

Gap	Impact
No Vapi integration	Phone calls don't work
No dashboard auth	Anyone with URL sees data
Single hardcoded tenant	Not multi-clinic SaaS
Calendar disabled	Booking not E2E verified
Browser-only webhooks	Phone can't use open-tab delivery
No Stripe billing	Can't charge customers
No HIPAA production stack	Can't handle real PHI at scale
Overall	~33% toward sellable product

3 Dashboards: Two Products, Not One

3.1 Critical Distinction

Super-Admin (yours) All clinics, MRR, churn Usage minutes, margins Onboard tenants System health (all) Support / failed webhooks <i>Not built yet</i>	Clinic Dashboard (client) Their leads & sessions Their emergencies Their config (edit) Their billing / usage Call recordings <i>Prototype exists</i>
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3.2 Clinic Dashboard — What to Display

Screen	Shows	Status
Overview	Calls today, leads, avg duration, emergencies	Built
Leads	Name, phone, DOB, insurance, reason	Built
Sessions	Full transcripts	Built
Emergency Log	Red-flag events	Built
Appointments	Booked slots, confirmations	Missing
Call Recordings	Audio playback	Missing
Configuration	Greeting, services, hours	Read-only
Usage & Billing	Minutes vs. plan, invoices	Missing

Clinic must NOT see: other clinics' data, your Vapi costs, margins, raw API keys.

3.3 Super-Admin Dashboard — What You Need

Screen
Tenants — all clinics, plan tier, onboarding stage
Revenue — MRR, ARR, churn, pilot pricing
Usage — minutes/tenant, overage, cost vs. revenue
System Health — Vapi/Twilio uptime, errors per tenant
Onboarding — wizard progress per clinic
Support — failed webhooks, stuck calls

4 Database: Current vs. Target

4.1 Existing Tables (SQLite)

Table	Purpose
<code>sessions</code>	One conversation; <code>client_id</code> , duration, flags
<code>conversation_turns</code>	Each message; role, intent, state
<code>leads</code>	Patient info: name, DOB, insurance, reason
<code>emergency_events</code>	Red-flag keyword detections
<code>audit_logs</code>	HIPAA trail (exists; barely used)

Config lives in `configs/*.json`, not the database. No `users`, `tenants`, or `appointments` tables yet.

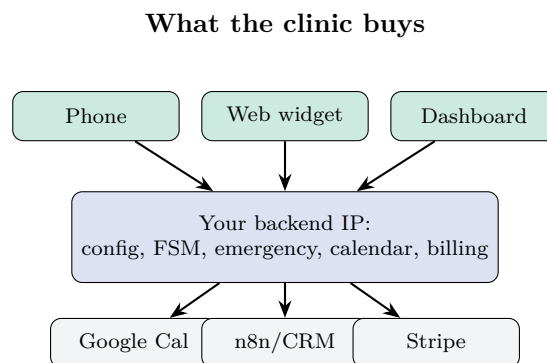
4.2 Target Schema (Production SaaS)

New Table	Purpose
<code>tenants</code>	Company, plan, Stripe ID, status
<code>users</code>	Email, role (<code>super_admin</code> / <code>clinic_admin</code> / <code>staff</code>)
<code>tenant_config</code>	Prompt, voice, hours, services (replaces JSON)
<code>phone_numbers</code>	E.164 → tenant mapping
<code>appointments</code>	Booked slots, Google event ID
<code>usage_records</code>	Minutes per tenant per month (billing)
<code>webhook_deliveries</code>	Outbound event status, retries
<code>patients</code>	Returning patient lookup (Phase 2)

4.3 Database Rules

- Every query filtered by `tenant_id`
- PHI encrypted at rest in production
- Audit log on every PHI read/write
- Sessions immutable after close
- PostgreSQL migration before 10+ clinics

5 What You Are Selling



Your moat: medical conversation design, emergency handling, tenant config, dashboard, compliance — not Vapi.

6 Pre-Code Checklist

6.1 Phase 0 — Align (Now)

Product definition agreed

Pick first pilot clinic (number, calendar, services)

Open accounts: Vapi, Twilio, Deepgram, OpenAI, ElevenLabs, GCP

6.2 Phase 1 — Phone MVP (~2 weeks)

1. `server/routes/vapi.js` — assistant-request, tool-calls, end-of-call
2. Phone number → `client_id` mapping
3. Wire Vapi tools → calendar, leads, emergency routes
4. Server-side webhook worker (replace browser outbox for phone)
5. Enable calendar; E2E booking test
6. Store `recording_url`, `channel: phone` on sessions

Exit: Call → book → transcript in dashboard in 30s.

6.3 Phase 2 — Clinic Dashboard v1

7. `users` + `tenants` tables; JWT auth
8. API routes require auth + tenant from token
9. Appointments panel; call recording playback
10. Editable config (greeting, hours, services)
11. Emergency email/push notification
12. Move system health internals to super-admin only

6.4 Phase 3 — First Paying Client (~weeks 3–6)

13. HIPAA hosting + BAAs (all providers)
14. Wire `audit_logs`; encrypt PHI at rest
15. Stripe tiers + `usage_records`
16. Production deploy; after-hours flow; call transfer; SMS confirm

Exit: One clinic pays \$999–1,499/mo; signed BAA; 2 weeks stable.

6.5 Phase 4 — Multi-Tenant SaaS (months 2–4)

19. Super-admin dashboard (/admin/*)
20. Tenant onboarding wizard
21. Config JSON → database
22. PostgreSQL; reschedule/cancel tools; CI/CD; monitoring

Exit: New clinic live in <2 hrs; 10 paying tenants.

6.6 Phase 5 — Scale (months 4–18)

Analytics, outbound campaigns, CRM connectors, EHR (Epic/athena), patient DB, white-label, LiveKit cost migration, SOC 2.

7 Decisions to Make Before Coding

Decision	Recommendation
Dashboard apps	One React app: /admin/* vs /clinic/*
Config storage	JSON for pilot; DB at L3
Production LLM	GPT-4o mini (phone); Ollama dev only
Twilio numbers	You own; include in subscription
Pricing	Flat tier + overage minutes
PHI transcripts	Store encrypted with BAA

8 60-Second Pitch

“MedVoice is an AI receptionist built for medical clinics. When a patient calls or chats, our assistant Aria answers immediately — even at 2am — collects their information, checks the clinic’s Google Calendar, and books the appointment. If someone says ‘chest pain,’ we use a hard-coded emergency system that logs the event and offers transfer to staff. Everything appears in a dashboard the office manager uses each morning: new leads, transcripts, and emergency alerts. Clinics pay \$1,500–3,000/month — less than half one receptionist — and never miss an after-hours call.”

9 Recommended Build Order

1. Pilot clinic — 2. Vapi phone — 3. Auth — 4. Clinic dash — 5. Super-admin → 6. Stripe + HIPAA